Expressions and Two-Step Equations

Concept

An expression has a value. An equation states that two expressions are equal.

Substitute

Replace variables using brackets: $x = -3 \to 2(-3)^2$.

Like terms

Combine matching variable parts: $3x + 5x = 8x$.

Balance

Do the same operation to both sides of an equation.

Two steps

In $ax + b = c$, undo b, then divide by $a \neq 0$.

Example 1

Solve $4x - 9 = 19$.

Solution

1. Add 9 to both sides: $4x = 28$.
2. Divide by 4: $x = 7$.
3. Check: $4(7) - 9 = 19$.

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Example 2

A rental costs $8 plus $6 per hour. Total: $38. Find the hours.

Solution

1. Let h = hours. Write $6h + 8 = 38$.

2. $6h = 30 \rightarrow h = 5$ hours.

3. Check: $6(5) + 8 = \$38$.

Expressions and Two-Step Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Evaluate $2p^2 + q$ for $p = -4$ and $q = 3$.

2. Solve $5x + 6 = 41$.

3. Solve $-3x + 4 = 22$.

4. Solve $x/4 - 7 = -2$.

5. Three identical tickets and a $4 fee cost $31. Find the price per ticket.

Expressions and Two-Step Equations

Practice answers | check after trying

1. $2(16) + 3 = 35$.

2. $5x = 35$, so $x = 7$.

3. $-3x = 18$, so $x = -6$.

4. $x/4 = 5$, so $x = 20$.

5. Let t be ticket price. $3t + 4 = 31$ gives $t = 9$. Each ticket costs $9; $3(9) + 4 = 31$.